PES UNIVERSITY | DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Lab 3: Component Modelling & Architectural Pattern Selection

Software Engineering Lab (UE24CS252AA) — Architectural Justification Document

Student: Sushruth S **SRN:** PES1UG24CS486 **Assigned System:** Coffee Kiosk **Style:** Layered Architecture

Architecture Selection: "We chose Layered Architecture for the Self-Service Coffee Kiosk System."

1. Architectural Choice:

We selected the classic 3-Tier Layered Architecture consisting of a Presentation Layer (Touch Screen User Interface Component), a Business Layer (Order Manager, Payment Service, and Receipt Printer Components), and a Data Layer (Database Component). Each layer encapsulates discrete responsibilities with strict top-down dependency flow, allowing components to interact solely through well-defined UML provided and required interfaces.

2. Two Reasons for Architectural Choice (Scenario-Related):

• **Reason 1 — Modularity & Hardware Decoupling in an Embedded Kiosk:** The kiosk interacts with specialized peripheral hardware (capacitive touch screen, EMV credit card reader terminal, and thermal receipt printer). A Layered style isolates these physical device interactions within modular business/hardware components. For example, if the receipt printer model or touch screen OS changes, only the respective driver component needs modification while core order orchestration and database schemas remain unaffected.

• **Reason 2 — Elimination of Unnecessary Distributed Complexity:** Unlike web-scale distributed e-commerce apps, a café coffee kiosk runs as a localized single-station embedded terminal serving sequential in-person orders. Adopting a Microservices architecture would introduce severe networking latency, serialization overhead, and multi-service failure states at checkout. Layered architecture guarantees near-instant in-process IPC method invocation, deterministic execution order, and trivial deployment on dedicated kiosk hardware.

3. Security Advantage:

Strict Layer Isolation & PCI-DSS Cardholder Protection: Because customers can pay exclusively via credit card, isolating payment logic inside an encapsulated Payment Service Component protects sensitive payment flows. The Presentation Layer (touch screen) has zero direct visibility or direct database access to stored transactional records or credit card memory buffers. All card transactions are funneled through the Payment Interface, which handles point-to-point encryption and tokenization with the card reader hardware before notifying the Order Manager, minimizing PCI compliance scope and preventing memory-scraping attacks on the public kiosk.

4. Performance Benefit:

Zero-Network Overhead & Local In-Memory Query Optimization: In a busy café rush, low latency is critical to prevent customer queues. In this Layered design, components reside in the same runtime host and interact through ultra-fast, local compiled/IPC interfaces rather than REST/gRPC HTTP network round-trips. Furthermore, static menu data (Espresso, Americano, Latte) and size pricing (Small, Large) are pre-loaded from the Database Component into fast local memory caches upon system boot, enabling instantaneous touch-screen rendering (<10ms) and swift end-to-end checkout completion.

Summary of System Components (5) & Required/Provided Interfaces (4):

Component Name	Assigned Layer	Interfaces (Ball / Socket)	Functional Responsibility
User Interface	Presentation	Requires: Order Interface (Socket)	Touch screen display, drink & size selection, status rendering
Order Manager	Business	Provides: Order Interface (Ball) Requires: Payment, Receipt, DB (Sockets)	Central orchestrator; computes prices; coordinates pay & print
Payment Service	Business	Provides: Payment Interface (Ball)	Card terminal hardware bridge, PCI encryption, card-only billing
Receipt Printer	Business (HW)	Provides: Receipt Interface (Ball)	ESC/POS driver interface, formats receipt slip, paper checks
Database	Data Layer	Provides: Database Interface (Ball)	Menu store (Espresso, Americano, Latte), pricing & order logs